Deep Field Reconstruction via Continuous Spatial-Spectral Fields: Problem Formulation, Epistemic Assumptions, and Statistical Claim Gates

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https://github.com/astroai/torchsky

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Abstract

We present a closed mathematical, statistical, and computational formulation for recovering optimal, ultra-deep astronomical fields from multi-epoch, multi-band imaging. The core epistemic premise is that no post-processing algorithm or neural prior can exceed the intrinsic Fisher information content of the native detector photon arrivals. However, by replacing classical lossy pipelines (resampling → stacking → detection → catalog photometry) with direct scene inference under the joint native-pixel likelihood, the method can substantially outperform existing practical estimators. We formulate sky reconstruction as a continuous inverse problem on $(\alpha, \delta, \lambda) \in \mathcal{S}^2 \times \mathbb{R}^+$, representing the non-negative latent field $I_\theta(\boldsymbol{x}, \lambda) \geq 0$ via continuous parameterizations (INRs, Riemannian tangent Gaussian Splats, and spherical Power Foams) and decomposing its projection into an identifiable passband subspace and an orthogonal, prior-regularized null space. We formulate a flexible mechanistic forward model in canonical detector electron counts $[e^-]$ that preserves geometric, radiometric, and optical invariants while learning gauge-closed deviations in passband transmission, non-stationary chromatic point-spread functions (PSFs), sub-pixel astrometric poses, time-dependent astronomical foregrounds, and instrumental rates. We catalogue a systematic inventory of 51 principal assumptions underlying the methodology, analyze population-specific expected headroom across distinct astrophysical source classes, establish an injection protocol protected against the inverse crime, and formulate rigorous statistical claim gates based on 50% completeness depth (m_{50}) at fixed false-positive surface density, matched effective resolution, and a two-stage out-of-sample held-out exposure photon prediction test.

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1 The Inverse Problem & Flexible Forward Formulation

1.1 The Continuous Latent Scene & Spectral Decomposition

Classical coaddition approaches resample discrete exposures onto a common pre-determined pixel grid and average them. This introduces interpolation covariance, irreversible sub-pixel information loss, and rigid per-band discretization. In contrast, we formulate the true physical sky as a continuous specific-intensity field on the celestial sphere:

$$I(\boldsymbol{x}, \lambda) \geq 0, \quad \boldsymbol{x} \in \mathcal{S}^2, \quad \lambda \in [\lambda_{\min}, \lambda_{\max}], \tag{1}$$

expressed in spectral photon radiance units $[\text{photons}\,\text{s}^{-1}\,\text{m}^{-2}\,\text{Å}^{-1}\,\text{sr}^{-1}]$. Energy-based conventions relate through the wavelength/frequency Jacobian and photon energy,

$$I_\nu(\boldsymbol{x}) = I_\lambda(\boldsymbol{x})\,\frac{c}{\lambda^2}, \qquad N_\lambda(\boldsymbol{x}, \lambda) = \frac{I_\lambda(\boldsymbol{x}, \lambda)\,\lambda}{hc}, \tag{2}$$

where I_ν is the familiar $\mathrm{MJy\,sr^{-1}}$ surface-brightness unit ($1\,\mathrm{Jy} = 10^{-26}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$), $I_\lambda = -\mathrm{d}I/\mathrm{d}\lambda$ is the energy spectral radiance per wavelength, and N_λ is the photon radiance entering the Poisson count statistics above.

The Passband Projection & Orthogonal Spectral Decomposition: Let B be the number of distinct broadband filters with spectral throughput curves $\{R_b(\lambda)\}_{b=1}^{B}$. The observation operator projects the continuous spectral dimension into a B-dimensional subspace:

$$\mathcal{R}_b[I](\boldsymbol{x}) = \int R_b(\lambda)\, I(\boldsymbol{x}, \lambda)\, \mathrm{d}\lambda, \quad b \in \{1, \ldots, B\}. \tag{3}$$

Because the filters are finite and broad, the linear mapping $\mathcal{R} : L_2([\lambda_{\min}, \lambda_{\max}]) \to \mathbb{R}^B$ has an infinite-dimensional spectral null space:

$$\ker(\mathcal{R}) = \left\{ \Delta I(\boldsymbol{x}, \lambda) : \int R_b(\lambda)\, \Delta I(\boldsymbol{x}, \lambda)\, \mathrm{d}\lambda = 0,\ \forall b \in \{1, \ldots, B\} \right\}. \tag{4}$$

The subspace $\mathcal{S}_{\mathrm{obs}} = \ker(\mathcal{R})^\perp = \mathrm{span}\{R_1, \ldots, R_B\}$ and null space $\ker(\mathcal{R})$ define a conservative *passband-only* spectral decomposition ($K_\lambda \leq B$). Modes in $\ker(\mathcal{R})$ are treated as prior-dominated and are not claimed as spectroscopically identified, although the full chromatic imaging operator may weakly constrain them through wavelength-dependent spatial response $K_n(\boldsymbol{u} \mid \boldsymbol{x}, \lambda)$.

To guarantee mathematical consistency between physical non-negativity and linear spectral decomposition, **the non-negativity constraint $I_\theta(\boldsymbol{x}, \lambda) \geq 0$ is applied to the total latent continuous field**. The observable and unobservable components are then defined rigorously by orthogonal projection:

$$I_{\mathrm{obs}}(\boldsymbol{x}, \lambda) \equiv \mathcal{P}_{\mathcal{S}_{\mathrm{obs}}}[I_\theta](\boldsymbol{x}, \lambda) = \sum_{k=1}^{K_\lambda} \langle I_\theta(\boldsymbol{x}, \cdot), \phi_k \rangle_{L_2}\, \phi_k(\lambda) \in \mathcal{S}_{\mathrm{obs}}, \tag{5}$$

$$I_{\mathrm{null}}(\boldsymbol{x}, \lambda) \equiv \mathcal{P}_{\ker(\mathcal{R})}[I_\theta](\boldsymbol{x}, \lambda) = I_\theta(\boldsymbol{x}, \lambda) - I_{\mathrm{obs}}(\boldsymbol{x}, \lambda) \in \ker(\mathcal{R}), \tag{6}$$

where $\{\phi_k(\lambda)\}_{k=1}^{K_\lambda}$ is an orthonormal basis spanning $\mathcal{S}_{\mathrm{obs}}$. While the total field is non-negative ($I_\theta \geq 0$), its linear projections I_{obs} and I_{null} are mathematically well-defined signed fields in L_2. Broadband observations directly constrain I_{obs}, whereas the unconstrained null-space modes I_{null} are regularized by an explicit astrophysical SED prior manifold:

$$\mathcal{R}_{\mathrm{SED}}\Big(\mathcal{P}_{\ker(\mathcal{R})}[I_\theta] \;\Big|\; \mathcal{P}_{\mathcal{S}_{\mathrm{obs}}}[I_\theta]\Big). \tag{7}$$

This ensures that all continuous spectral variation beyond $\mathcal{S}_{\mathrm{obs}}$ is formally an inductive prior completion, not an unconstrained observational measurement.

1.2 Epistemic Principles & Information-Theoretic Bounds

A foundational principle governs this entire formulation:

> **The Epistemic Ceiling:** *No mathematical representation, neural prior, or optimization method can create Fisher information beyond that contained in the raw native detector electron counts $\boldsymbol{y} = (y_{11}, \ldots, y_{Np})$. What the method can outperform is every practical estimator that compresses or discards this information through intermediate processing stages.*

Conventional astronomy pipelines follow a cascade of intermediate compressions:

$$\text{Raw Photons } \boldsymbol{y} \xrightarrow{\text{Resample}} \text{Interpolated Frames} \xrightarrow{\text{Coaddition}} \text{Static Stacks } \boldsymbol{z} \xrightarrow{\text{Filtering}} \text{Detections} \xrightarrow{\text{Apertures}} \text{Photometry}.$$

In contrast, our approach directly estimates the latent scene by solving, once and for all abstract scene representation $\theta \in \Theta_{\text{rep}}$ and abstract operator nuisance bundle $\psi \in \Psi$:

$$\hat{I}_\theta, \hat{\psi} = \arg \min_{\theta \in \Theta_{\text{rep}}, \psi \in \Psi} \left[-\sum_{n=1}^{N} \log p(\boldsymbol{y}_n \mid \mathcal{A}_n(\psi)[I_\theta]) + \mathcal{R}(\theta) + \mathcal{C}(\psi) \right]. \tag{8}$$

The objective is deliberately representation-agnostic: Θ_{rep} instantiates to the INR, Gaussian-splat, or power-foam parameterizations of §1.3, and Ψ instantiates to any rung of the hybrid operator ladder of §1.6 (frozen analytic, fitted parametric, or learned low-rank). All likelihoods (§1.7), calibration gauges (§1.5), and claim gates (§5) are stated at this abstract (θ, ψ) level; no gate depends on which scene representation or operator rung produced a candidate product. The method approaches the ultimate physical information ceiling by exploiting four legitimate physical and statistical mechanisms:

1. **Native-Pixel Geometry & Dither Phase Retention:** Retaining each individual exposure's WCS, sub-pixel dither offset (e.g., 0.03 pixel shifts), chromatic PSF, and noise plane avoids the blurring and correlated noise of raster coaddition.

2. **Pre-Detection Cross-Band Evidence Accumulation:** If an astrophysical source satisfies a plausible SED prior manifold $\mathcal{M}_{\text{SED}}$, weak signals across multiple passbands combine coherently before catalog thresholding:

$$\text{SNR}^2_{\text{joint}} \simeq \sum_{b=1}^{B} \text{SNR}^2_b. \tag{9}$$

   For example, six sub-threshold 1.5σ measurements can yield nominal 3.7σ joint evidence ($\sqrt{6} \times 1.5 = 3.67$) under an independently specified SED hypothesis, before accounting for trials over SED parameter space.

3. **Variance Reduction via Principled Astrophysical Priors:** By restricting the search space from all unconstrained images to plausible physical astrophysical scenes $I \in \mathcal{M}_{\text{plausible}}$ (spatial compactness, smooth SEDs, high sparsity in empty sky), the estimator can substantially reduce mean squared error when the prior is sufficiently well specified:

$$\text{MSE}(\hat{I}_{\text{prior}}) < \text{MSE}(\hat{I}_{\text{weak-prior}}). \tag{10}$$

1.3 Continuous-Domain Spatial-Spectral Parameterizations

We formulate three continuous-domain representations of $I_\theta(\boldsymbol{x}, \lambda)$, ensuring geometric validity on $\mathcal{S}^2$ and physical non-negativity:

1. **Implicit Neural Representations (INRs):** The field is parameterized by a continuous coordinate encoder $E(\boldsymbol{x}, \lambda)$ (multi-resolution spatial-spectral hash grids, Fourier features, or tensor planes) and a compact neural decoder f_θ, with non-negativity enforced by a strictly positive activation:

$$I_\theta(\boldsymbol{x}, \lambda) = \text{Softplus}\Big(f_\theta\big(E(\boldsymbol{x}, \lambda)\big)\Big) \geq 0. \tag{11}$$

2. **3D Gaussian Splats on Riemannian Tangent Charts:** To maintain geometric validity on $\mathcal{S}^2$, spatial centroids are defined in a local Riemannian normal coordinate chart $\boldsymbol{\xi} = \log_{\boldsymbol{x}_0}(\boldsymbol{x}) \in \mathbb{R}^2$ centered at reference sky position $\boldsymbol{x}_0$, combined with a normalized spectral coordinate $w = (\lambda - \lambda_0)/\Delta\lambda$:

$$I_\theta(\boldsymbol{\xi}, w) = \sum_{k=1}^{K} A_k \exp\left(-\frac{1}{2} (\boldsymbol{v} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1} (\boldsymbol{v} - \boldsymbol{\mu}_k) \right), \quad \boldsymbol{v} = (\boldsymbol{\xi}^\top, w)^\top, \tag{12}$$

where $A_k = \mathrm{Softplus}(\tilde{A}_k) \geq 0$ guarantees non-negativity, and $\boldsymbol{\Sigma}_k \succ 0$ is a positive-definite 3D covariance. In factorized form, spatial 2D Gaussians $\mathcal{N}(\boldsymbol{\xi}; \boldsymbol{\mu}_k^{\mathrm{sky}}, \boldsymbol{\Sigma}_k^{\mathrm{sky}})$ are modulated by strictly non-negative SEDs:

$$\mathrm{SED}_k(\lambda) = \mathrm{Softplus}\left(\sum_{m=1}^{K_\lambda} c_{km}\phi_m(\lambda)\right) \geq 0. \tag{13}$$

3. **Piecewise-Affine Spectral Power Foams on $\mathcal{S}^2$:** The field is partitioned into geometric cells $\{C_j\}$ defined by a bounded power diagram on $\mathcal{S}^2$ with sites $\boldsymbol{s}_j \in \mathcal{S}^2$ and radii r_j. Using the Riemannian logarithmic map $\boldsymbol{\xi}_j(\boldsymbol{x}) = \log_{\boldsymbol{s}_j}(\boldsymbol{x}) \in T_{\boldsymbol{s}_j}\mathcal{S}^2 \cong \mathbb{R}^2$, the field is a continuous-domain piecewise-affine representation:

$$I_\theta(\boldsymbol{x}, \lambda) = \sum_j \mathbf{1}_{C_j}(\boldsymbol{x}) \left[\sum_{k=1}^{K_\lambda} \left(c_{jk} + \boldsymbol{h}_{jk}^\top \log_{\boldsymbol{s}_j}(\boldsymbol{x})\right)\phi_k(\lambda)\right]_+, \tag{14}$$

where $[\cdot]_+ = \max(\cdot, 0)$ guarantees non-negativity. While the hard field is piecewise discontinuous across cell boundaries, geometric site optimization employs a differentiable annealed soft-surrogate ownership weight $\widetilde{w}_j(\boldsymbol{x}; \tau)$, whereas final scientific evaluation uses exact hard polygon clipping.

Representation Contract & Model Zoo: Every scene representation admitted into Θ_{rep} — implemented in torchsky as the `FieldModelKind` registry (24 kinds spanning INR, splat, foam, and feature-field families) — must satisfy a single uniform contract so that likelihoods, gauges, and gates remain representation-agnostic:

1. **Differentiable render with JVP/VJP:** the map $\theta \mapsto$ rendered native-pixel rate cube exposes exact Jacobian-vector and vector-Jacobian products for gradient-based joint optimization;
2. **Non-negativity:** rendered specific intensity satisfies $I_\theta(\boldsymbol{x}, \lambda) \geq 0$ everywhere;
3. **Shared orthonormal passband basis:** all kinds express spectra in the same orthonormal passband basis (the shared whitening transform of `spectral_basis_whitening_transform`), making cross-kind held-out likelihoods directly comparable;
4. **Exact final rendering:** scientific products use the exact (hard, non-surrogate) rendering path — no annealing approximations survive to evaluation.

Benchmark scoring loops over this zoo uniformly; model selection (§5) treats every (scene kind, operator rung) cell identically.

1.4 Rigorous Forward Model & Expected Electron Counts

Let $n \in \{1, \ldots, N\}$ index exposures, and let $\boldsymbol{y}_n$ denote the instrument-native measurement object (integrated pixel counts, calibrated slope/rate images, or non-destructive group read cubes).

We formulate the total incident celestial radiation field for exposure n as:

$$I_n(\boldsymbol{x}, \lambda) = I_{\mathrm{target}}(\boldsymbol{x}, \lambda) + I_{\mathrm{fg},n}(\boldsymbol{x}, \lambda), \tag{15}$$

where $I_{\mathrm{target}}(\boldsymbol{x}, \lambda)$ is the static celestial target scene, and $I_{\mathrm{fg},n}(\boldsymbol{x}, \lambda)$ is a constrained low-dimensional exposure-dependent astronomical foreground (e.g., time-variable zodiacal light, airglow, and scattered moonlight) parameterized as a smooth spatial polynomial modulated by known foreground SED templates.

The underlying expected integrated electron count Λ_{np} in pixel p is:

$$\Lambda_{np}(\theta,\psi) = \Lambda_{np}^{\mathrm{dark}}(\psi) + \Lambda_{np}^{\mathrm{thermal}}(\psi) + \int_{\Omega_{np}^{\mathrm{det}}} \mathrm{d}^2\boldsymbol{u} \int \mathrm{d}\lambda\, \mathcal{T}_n(\lambda,\boldsymbol{u};\psi) \times \int_{\mathcal{S}^2} \mathrm{d}\Omega_{\boldsymbol{x}}\, K_n(\boldsymbol{u} \mid \boldsymbol{x},\lambda,t_n;\psi)\, I_n(\boldsymbol{x},\lambda), \tag{16}$$

where:

- $\Omega_{np}^{\mathrm{det}} \subset \mathbb{R}^2$ is the continuous geometric footprint of detector pixel p in focal-plane coordinates $\boldsymbol{u} = (u_x, u_y)$.
- $\mathcal{T}_n(\lambda,\boldsymbol{u};\psi) = A_{\mathrm{eff}}\, t_n\, R_n(\lambda,\boldsymbol{u};\psi)$ is the total conversion throughput in $[e^-\,\mathrm{m}^2\,\mathrm{s\,photons}^{-1}]$, where A_{eff} is telescope collecting area, t_n is exposure duration, and $R_n(\lambda,\boldsymbol{u})$ is the dimensionless filter/detector quantum efficiency.
- $K_n(\boldsymbol{u} \mid \boldsymbol{x},\lambda,t_n;\psi)$ is the true non-stationary spatial response / PSF kernel mapping incident sky direction $\boldsymbol{x} \in \mathcal{S}^2$ to continuous detector coordinate $\boldsymbol{u}$. It satisfies the spatial normalization:

  $$\int_{\mathbb{R}^2} K_n(\boldsymbol{u} \mid \boldsymbol{x},\lambda,t_n;\psi)\, \mathrm{d}^2\boldsymbol{u} = 1, \quad \forall \boldsymbol{x} \in \mathcal{S}^2,\ \lambda \in \mathbb{R}^+,\ t_n. \tag{17}$$

  For a distorted WCS $W_n : \boldsymbol{u} \mapsto \boldsymbol{x}$ with differentiable astrometric pose residual $\boldsymbol{\delta}_n(\boldsymbol{u}) = \mathrm{Pose}_{\phi_n}(\boldsymbol{u})$, the kernel specializes locally to:

  $$K_n(\boldsymbol{u} \mid \boldsymbol{x},\lambda,t_n) \approx \mathcal{K}_n\Big(\boldsymbol{u} - W_n^{-1}(\boldsymbol{x}) - \boldsymbol{\delta}_n(\boldsymbol{u});\, \boldsymbol{u},\lambda,t_n\Big). \tag{18}$$

- $\Lambda_{np}^{\mathrm{dark}}(\psi)$ and $\Lambda_{np}^{\mathrm{thermal}}(\psi)$ are expected instrumental dark current and thermal counts.

For single-integration count data, the detected pixel value y_{np} is generated by integer Poisson arrivals $k_{np} \sim \mathrm{Pois}(\Lambda_{np})$ combined with deterministic electronic bias b_{np} and zero-mean Gaussian readout noise $\epsilon_{\mathrm{read},np} \sim \mathcal{N}(0, \sigma^2_{\mathrm{read},np})$:

$$y_{np} = b_{np} + k_{np} + \epsilon_{\mathrm{read},np}. \tag{19}$$

For calibrated pipeline products where electronic bias has already been subtracted, $b_{np} = 0$.

1.5 Closure of Calibration and Nuisance Gauges

To eliminate unidentifiable parameter degeneracies, we impose explicit, closed gauge constraints:

1. **Photometric Gauge ($I \leftrightarrow \mathcal{T}_n$):** Absolute filter throughput shapes $R_n^0(\lambda)$ and photometric zero-points are externally calibrated and fixed from standard star observations. The learnable spatially varying transmission residual $\Delta r_n(\lambda,\boldsymbol{u})$ in $R_n(\lambda,\boldsymbol{u}) = R_n^0(\lambda)\exp(\Delta r_n(\lambda,\boldsymbol{u}))$ is constrained to zero focal-plane spatial mean:

   $$\int_{\Omega^{\mathrm{det}}} \Delta r_n(\lambda,\boldsymbol{u})\, \mathrm{d}^2\boldsymbol{u} = 0, \quad \|\Delta r_n\|_\infty \le \epsilon_R, \tag{20}$$

   and anchored by calibration stars and flat-field likelihood terms $\mathcal{C}_{\mathrm{filter}}(\psi)$.

2. **Astrometric Gauge ($\boldsymbol{\delta}_n \leftrightarrow$ PSF Centroid):** One reference exposure is frozen ($\boldsymbol{\delta}_1(\boldsymbol{u}) \equiv \mathbf{0}$) to fix the global astrometric chart. The spatial response kernel is strictly zero-centroid:

   $$\int_{\mathbb{R}^2} \Delta\boldsymbol{u}\, \mathcal{K}_n(\Delta\boldsymbol{u};\boldsymbol{u},\lambda,t_n)\, \mathrm{d}^2\Delta\boldsymbol{u} = \mathbf{0}, \quad \forall \boldsymbol{u},\lambda,t_n. \tag{21}$$

3. **Target Diffuse Sky vs. Astronomical Foreground Gauge ($I_{\text{target}}^{\text{diffuse}} \leftrightarrow I_{\text{fg},n}$):** To break the degeneracy between the static diffuse celestial scene $I_{\text{target}}^{\text{diffuse}}$ and time-dependent astronomical foregrounds $I_{\text{fg},n}$, we decompose the foreground into an externally anchored baseline plus zero-mean exposure variations:

$$I_{\text{fg},n}(\boldsymbol{x}, \lambda) = I_{\text{fg},0}(\boldsymbol{x}, \lambda) + \Delta I_{\text{fg},n}(\boldsymbol{x}, \lambda), \quad \sum_n w_n \, \Delta I_{\text{fg},n}(\boldsymbol{x}, \lambda) = 0, \tag{22}$$

where $I_{\text{fg},0}$ is fixed from physical zodiacal/airglow models (or I_{target} is defined relative to the exposure-weighted mean sky foreground), and instrumental Poisson counts $\Lambda_{np}^{\text{dark}}, \Lambda_{np}^{\text{thermal}}$ and electronic bias b_{np} are identified strictly from dark/shuttered calibration frames.

1.6 Hybrid Operator Models: A Ladder of Learned and Not-Learned Nuisance Fields

No forward-model component may be hard-wired to a single analytic form. Every operator slot in ψ — PSF, background, structured noise — is a latent field with its own prior and gauge, estimated at the run's chosen rung of a three-rung ladder:

1. `frozen_analytic`**:** physics-pinned constants (e.g., a tabulated diffraction-limited kernel); no free parameters, no calibration data required.
2. `fitted_parametric`**:** low-dimensional parametric forms (per-band Gaussian σ, polynomial background) fitted offline from calibration evidence.
3. `learned_lowrank`**:** neural fields over focal-plane position $\boldsymbol{u}$ and wavelength λ, anchored by calibration data and regularized toward smoothness.

The learned PSF is formalized as a spatially and spectrally varying Gaussian family

$$K_n(\Delta\boldsymbol{u} \mid \boldsymbol{u}, \lambda) = \mathcal{G}\big(\Delta\boldsymbol{u};\, \boldsymbol{\phi}(\boldsymbol{u}, \lambda)\big), \tag{23}$$

with kernel parameters $\boldsymbol{\phi} = (\sigma_x, \sigma_y, \rho)$ produced by a bilinear spatial control-grid interpolant modulated by an in-band spectral Legendre basis; positivity is enforced by construction (softplus parameterization), and the existing gauges of §1.5 apply unchanged: unit integral $\int K \, \mathrm{d}^2\Delta\boldsymbol{u} = 1$ and zero centroid $\int \Delta\boldsymbol{u} \, K \, \mathrm{d}^2\Delta\boldsymbol{u} = \mathbf{0}$, plus a smoothness prior on $\boldsymbol{\phi}(\boldsymbol{u}, \lambda)$.

Identifiability Decision: Learned operator components are estimated *only* from calibration stars (and judged on held-out validation-star negative log-likelihood) — never jointly from scene residuals. This closes the degeneracy in which astrophysical sources are absorbed into a flexible PSF: a learned kernel that must explain unresolved structure would otherwise trade flux against the scene θ. The deliberate 5%-wider fitted Gaussian remains in the protocol as the PSF-misspecification control arm.

1.7 Instrument-Specific Native Detector Likelihoods

Rather than imposing a single universal noise distribution, the measurement likelihood is formally instrument-specific:

$$p(\boldsymbol{y}_n \mid I_\theta, \psi) = \mathcal{L}_n\Big(\boldsymbol{y}_n;\, \mathcal{A}_n(\psi)[I_\theta], \psi_n\Big), \tag{24}$$

structured according to the detector architecture and readout physics:

1. Canonical Single-Integrated-Count Likelihood (CCD / Single Integration): For calibrated single-read integrations, integer Poisson arrivals $k_{np} \sim \text{Pois}(\Lambda_{np})$ combine with deterministic bias b_{np} and independent Gaussian read noise $\epsilon_{\text{read},np} \sim \mathcal{N}(0, \sigma^2_{\text{read},np})$:

$$p(y_{np} \mid \Lambda_{np}, b_{np}) = \sum_{k=0}^{\infty} \frac{\Lambda_{np}^k e^{-\Lambda_{np}}}{k!} \frac{1}{\sqrt{2\pi\sigma^2_{\text{read},np}}} \exp\left(-\frac{(y_{np} - b_{np} - k)^2}{2\sigma^2_{\text{read},np}}\right). \tag{25}$$

For continuous gradient optimization in count space, the exact second-order variance $V_{np}(\Lambda_{np}) = \sigma^2_{\text{read},np} + \Lambda_{np}$ yields the valid heteroscedastic Gaussian approximation:

$$\mathcal{L}_{\text{PG}}(\theta, \psi) = \frac{1}{2} \sum_{n,p} m_{np} \left[\frac{(y_{np} - b_{np} - \Lambda_{np}(\theta, \psi))^2}{\sigma^2_{\text{read},np} + \Lambda_{np}(\theta, \psi)} + \log\left(2\pi(\sigma^2_{\text{read},np} + \Lambda_{np}(\theta, \psi))\right)\right]. \tag{26}$$

2. Calibrated Count-Rate Likelihood with Correlated $1/f$ Readout Covariance: For calibrated slope/rate products $\boldsymbol{y}_n$ (e.g., JWST NIRCam `rate.fits` or `cal.fits`), residual $1/f$ striping and amplifier correlated noise across rows/columns can inflate apparent detection significance near the noise floor if treated as independent. We formulate the correlated Gaussian likelihood:

$$\mathcal{L}_{\text{rate}}(\boldsymbol{y}_n;\, \boldsymbol{\mu}_n) = \frac{1}{2}(\boldsymbol{y}_n - \boldsymbol{\mu}_n)^\top \boldsymbol{\Sigma}_n^{-1}(\boldsymbol{y}_n - \boldsymbol{\mu}_n) + \frac{1}{2}\log\det\left(2\pi\boldsymbol{\Sigma}_n\right), \tag{27}$$

where $\boldsymbol{\Sigma}_n = \text{diag}(\boldsymbol{V}(\boldsymbol{\mu}_n)) + \boldsymbol{\Sigma}_{1/f}$ incorporates the spatial/temporal $1/f$ noise covariance $\boldsymbol{\Sigma}_{1/f}$ estimated from reference pixels or pre-whitening filters.

3. Truly Native Multiaccum Up-the-Ramp Likelihood (Non-Destructive Reads): For infrared arrays (JWST NIRCam/NIRISS, Roman WFI), the raw data consist of G non-destructive group samples $\{y_{n,p,g}\}_{g=1}^{G}$. The group increments $\Delta y_{n,p,g} = y_{n,p,g} - y_{n,p,g-1}$ represent incremental Poisson arrivals $k_{n,p,g} \sim \text{Pois}(\Lambda_{n,p,g})$ plus differenced read noise with structured inter-group covariance ($\text{Cov}(\Delta\epsilon_g, \Delta\epsilon_{g+1}) = -\sigma^2_{\text{read}}$), augmented by correlated $1/f$ and common-mode readout components. Operating directly on group-level increments allows joint cosmic-ray jump rejection and $1/f$ flicker mitigation coincident with latent scene reconstruction, honoring the truly native measurement physics.

4. Heavy-Tailed Robust Objectives for Outlier Immunity: To reject unmasked cosmic rays and satellite streaks without discarding valid faint sources, we incorporate pseudo-Huber/Charbonnier or Student-t robust loss functions:

$$\mathcal{L}_{\text{robust}}(\theta, \psi) = \sum_{n,p} m_{np}\, \rho\left(\frac{y_{np} - b_{np} - \Lambda_{np}(\theta, \psi)}{\sqrt{\sigma^2_{\text{read},np} + \Lambda_{np}}};\, \epsilon_{\text{robust}}\right), \quad \rho(z) = \sqrt{z^2 + \epsilon^2_{\text{robust}}} - \epsilon_{\text{robust}}. \tag{28}$$

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2 Systematic Assumption Inventory & Risk Analysis

The scientific integrity and operational validity of this formulation depend on an explicit stack of 52 principal physical, statistical, and operational assumptions. We catalogue these assumptions in Table 1, categorized by their nature and scientific criticality.

Table 1: Systematic inventory of 52 principal assumptions underlying continuous deep field reconstruction.

#	Assumption Description	Type	Scientific Criticality
A. Representation & Astrophysical Reality			
1	Target sky is a continuous non-negative field $I(\boldsymbol{x}, \lambda) \geq 0$.	Explicit	Fundamental representation
2	Scientifically interesting target scene is static across epochs.	Implicit	High (transients/movers excluded)
3	Exposure-dependent emission separates into low-dim foreground $I_{\mathrm{fg},n}$.	Explicit	High for ground & space data
4	Foregrounds are well described by smooth spatial polynomials & SEDs.	Explicit	Potential failure in complex cirrus
5	Parameterizations (INRs, Foams, Splats) have sufficient capacity.	Explicit	Critical representation gate
6	Representation inductive biases are beneficial near the noise floor.	Implicit	**Central research hypothesis**
7	Positivity is physically valid after foreground subtraction.	Explicit	Safe physical constraint
8	Local Riemannian tangent charts are geometrically accurate.	Explicit	Valid for survey cutouts ($\leq 2^\circ$)
9	Piecewise-affine cells adequately capture structured/diffuse scenes.	Explicit	Empirical modeling choice
10	Soft site optimization $\rightarrow$ hard evaluation has negligible mismatch.	Implicit	Validation checkpoint
B. Spectral Subspace & Prior Manifolds			
11	Broadband imaging robustly identifies finite subspace $\mathcal{S}_{\mathrm{obs}}$.	Explicit	Conservative identifiability
12	Modes in $\ker(\mathcal{R})$ are prior-dominated, not measured spectroscopy.	Explicit	Essential for scientific honesty
13	Weak sub-band chromatic PSF dispersion is conservatively not exploited.	Explicit	Methodological bound
14	Adopted prior manifold $\mathcal{M}_{\mathrm{SED}}$ contains realistic galaxy/star SEDs.	Implicit	**Critical for cross-band gains**
15	Prior manifold is broad enough not to suppress rare/exotic discoveries.	Implicit	**Critical for discovery claims**
16	Parameterization activations enforce total field positivity $I_\theta \geq 0$.	Explicit	Mathematical consistency
C. Instrument, Geometry & Calibration Gauges			
17	Image formation is strictly linear in incident optical photon flux.	Explicit	Fundamental optics
18	Non-linearities (saturation, persistence, IPC, brighter-fatter) are calibrated.	Implicit	Detector pipeline dependency
19	Distortion WCS mappings are locally differentiable and invertible.	Explicit	Standard astrometry
20	Residual astrometric misregistration is captured by $\boldsymbol{\delta}_n(\boldsymbol{u})$.	Implicit	Critical for sub-pixel sharpness
21	Non-stationary chromatic response is normalized: $\int K_n \mathrm{d}^2\boldsymbol{u} = 1$.	Explicit	Radiometric conservation
22	Sufficient stellar calibration exists to anchor PSF spatial variation.	Implicit	Calibration requirement
23	Throughput residuals Δr_n are smooth and zero-mean on focal plane.	Explicit	Photometric gauge closure
24	Absolute passbands and zero-points are externally calibrated.	Explicit	Critical absolute scale

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#	Assumption Description	Type	Scientific Criticality
25	Reference exposure freeze ($\boldsymbol{\delta}_1 \equiv \mathbf{0}$) fixes astrometric gauge.	Explicit	Astrometric gauge closure
26	Mean foreground $I_{\mathrm{fg},0}$ is anchored, closing target diffuse gauge.	Explicit	Celestial/foreground closure
27	Dark current Λ^{dark} and thermal rate $\Lambda^{\mathrm{thermal}}$ are known from darks.	Explicit	Detector calibration
28	Electronic bias b_{np} is calibrated or deterministically subtracted ($b = 0$).	Explicit	Readout calibration
29	Dither diversity provides sufficient spatial phase diversity.	Implicit	**Critical for super-resolution**
D. Statistical Likelihood & Noise Physics			
30	Photon arrivals and dark events obey Poisson statistics.	Explicit	Fundamental physics
31	Readout noise is independent, zero-mean Gaussian $\mathcal{N}(0, \sigma^2_{\mathrm{read}})$.	Explicit	Standard detector model
32	Correlated readout noise (e.g., $1/f$) is modeled or prewhitened.	Explicit	**Critical for JWST faint depth**
33	Cosmic rays and bad pixels are rejected by robust loss / mask m_{np}.	Explicit	Data cleaning integrity
34	Heteroscedastic Gaussian log-likelihood approximates count distribution.	Explicit	Computational tractability
35	Regularization hyperparameters are selected without test set leakage.	Implicit	Statistical validity
E. Optimization, Solvers & Numerical Scaling			
36	Non-convex joint optimization reaches a scientifically sound optimum.	Implicit	**Major practical engineering risk**
37	Alternating/IRLS optimization avoids degenerate nuisance/scene trade-offs.	Implicit	Convergence stability
38	Linear sub-solvers (Jacobi-PCG) converge without matrix materialization.	Explicit	Scalability requirement
39	Numerical quadrature errors remain well below photon noise floor.	Explicit	Discretization precision
40	Spatial-spectral tile streaming preserves global scene continuity.	Implicit	Large-scale execution
F. Benchmarking, Falsification & Claim Gates			
41	Native-pixel GLRT is a rigorous comparator for native information.	Explicit	Essential baseline fairness
42	Injections are rendered by independent simulators (no inverse crime).	Explicit	**Critical validation integrity**
43	Blank/null fields reproduce real empirical false-positive generation.	Implicit	False-discovery control
44	Injections span the full distribution of morphology, SED, and crowding.	Explicit	Completeness calibration
45	Fixed false-positive surface density ρ_{false} replaces sample purity.	Explicit	Prevalence independence
46	Matched effective spatial resolution $\mathcal{H}_{\mathrm{eff}}$ prevents smoothing cheats.	Explicit	**Guarantees true sensitivity**
47	Photometric flux ($\leq 2\%$) and color ($\leq 0.02\,\mathrm{mag}$) biases are bounded.	Explicit	Photometric integrity
48	Depth gains generalize across independent spatial-block cross-validation.	Explicit	Spatial robustness
49	Hyperparameters and detection thresholds are frozen before blind evaluation.	Explicit	Pre-registration integrity

Continued on next page

#	Assumption Description	Type	Scientific Criticality
50	Discovered sources predict photon arrivals in held-out test dithers.	Explicit	**Decisive falsification test**
51	All benchmark methods receive identical exposures, masks, and priors.	Explicit	**Benchmark fairness**
52	Learned operator components (ϕ) are fitted only on calibration stars and judged on held-out validation NLL — never jointly from scene residuals.	Explicit	**Identifiability: no source absorption into PSF**

The Central Risk: Prior-Model Mismatch vs. Hallucination: The central research hypothesis (Assumption 6) asserts that inductive biases near the noise floor legitimately reduce mean squared error by trading small bias for large variance reduction. However, this creates the primary risk of the methodology: *the network declaring that a noisy sub-threshold fluctuation looks like a high-redshift galaxy.* To prevent hallucinated discoveries from corrupting science catalogues, our statistical claim gates require that any candidate source discovered exclusively by the continuous model must pass empirical transfer-function validation and a two-stage out-of-sample held-out exposure photon prediction test.

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3 Modular Architecture & Operator Contracts in Torchsky

3.1 Linear Adjoints vs. Nonlinear Differentiable Primitives

Torchsky enforces an explicit mathematical contract across its operator graph:

- **Linear Measurement Primitives** (e.g., fixed-geometry pixel integration $\boldsymbol{F}_n$, fixed linear PSF convolution $\boldsymbol{H}_n$, and spectral basis projection $\boldsymbol{B}_n$) expose exact linear adjoint operators $\mathcal{A}^\dagger$ satisfying numerical adjoint dot-product tests:

$$\langle \mathcal{A}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \mathcal{A}^\dagger \boldsymbol{y} \rangle \quad \text{to machine precision } (< 10^{-7} \text{ in float32}). \tag{29}$$

- **Nonlinear Parameterized Primitives** (e.g., neural coordinate decoders f_θ, differentiable pose fields $\boldsymbol{\delta}_n$, adaptive foam site geometry $\boldsymbol{s}_j$, learnable transmission Δr_n) expose differentiable Jacobian-Vector Products (JVP) and Vector-Jacobian Products (VJP), whose localized Jacobians $\boldsymbol{J}$ pass linear adjoint tests:

$$\langle \boldsymbol{J}\boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{v}, \boldsymbol{J}^\top \boldsymbol{w} \rangle. \tag{30}$$

3.2 Matrix-Free Linear Subsolvers in Iterative Optimization

For fixed geometric representations (e.g., fixed-site Power Foams or linear Gaussian splat bases), the inner appearance subproblem solves the regularized normal equations:

$$\left(\sum_n \boldsymbol{F}_n^\top \boldsymbol{H}_n^\top \boldsymbol{W}_n \boldsymbol{H}_n \boldsymbol{F}_n + \gamma \boldsymbol{D}^\top \boldsymbol{D} \right) \boldsymbol{c} = \sum_n \boldsymbol{F}_n^\top \boldsymbol{H}_n^\top \boldsymbol{W}_n \boldsymbol{y}_n + \gamma \boldsymbol{D}^\top \boldsymbol{D} \boldsymbol{c}_0, \tag{31}$$

solved matrix-free via Preconditioned Conjugate Gradients (Jacobi-PCG). For robust, heteroscedastic, or nonlinear objectives, this linear solve is embedded as the inner step of an outer Iteratively Reweighted Least Squares (IRLS) or Gauss-Newton framework.

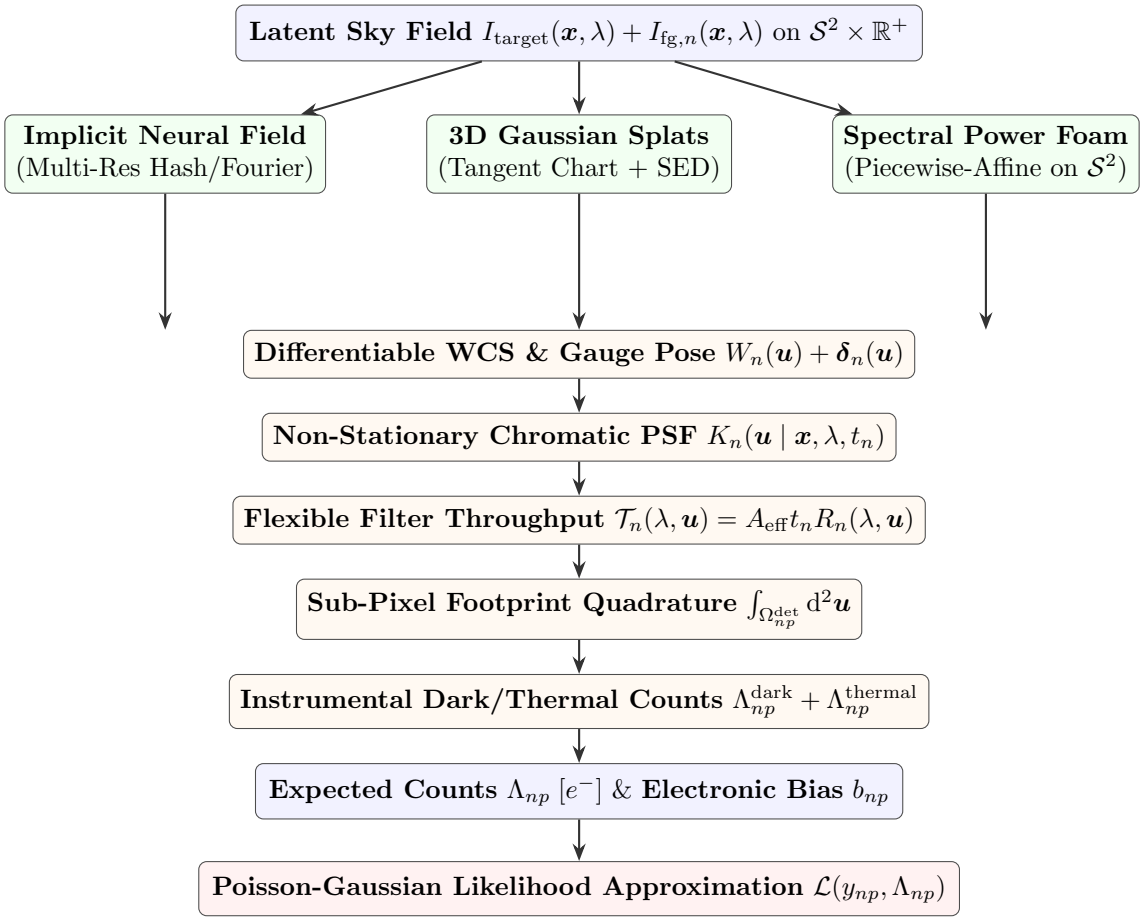


Figure 1: Modular, differentiable forward observation pipeline in `torchsky`.

4 Classical Baselines & Rigorous Experimental Protocol

4.1 Comprehensive Comparator Matrix

To prevent straw-man evaluations, learned continuous models are benchmarked against both classical coaddition pipelines and native-pixel un-resampled baselines, ensuring identical SED and source-family assumptions wherever feasible:

Pipeline / Family	Specific Configuration	Role in Benchmark
Torchsky Stack	Inverse-variance weighted bilinear/splat	Internal classical mapmaking control
Upstream Drizzle	`drizzlepac` square Drizzle (pixfrac $\in \{0.6, 0.8, 1.0\}$)	Standard space-telescope (HST/JWST) comparator
SWarp	SWarp v2.41.5 (`BILINEAR` & `LANCZOS3` with SIP$\rightarrow$TPV)	Standard ground/wide survey comparator
Montage	Area-weighted spherical polygon reprojection	Exact spherical geometry baseline
Whitened Multi-Band χ^2	Optimal matched filter over the best classical stacks	Strongest classical stack-level detection
Native-Pixel GLRT	Multi-band matched filter directly on all un-resampled exposures with matched SED/source prior	**Strongest native-pixel baseline**; isolates representation gains from stacking and prior effects
Linear Information Bound	Native-detector un-resampled Fisher linear estimate	Unbiased theoretical sensitivity reference

Table 2: Comprehensive comparator matrix executed on identical calibrated detector inputs and permissible priors.

4.2 Strict Protection Against the Inverse Crime

To ensure that scientific depth gains are genuine and not artifacts of model self-consistency, **synthetic injections are strictly decoupled from the inference forward operator**:

- **Independent Simulator for Truth Generation:** Injected truth sources are rendered using independent, physics-based optical wavefront simulators (e.g., WebbPSF utilizing measured optical path differences, pupil state models, and high-frequency jitter), empirical laboratory passbands with measured pupil angle variations, Gaia DR3 astrometric solutions, and realistic non-parametric galaxy morphologies from high-resolution HST/JWST cutouts.
- **Imperfect Inference Initialization:** The reconstruction engine initializes from nominal telescope models and must learn or fit residual calibrations without access to the simulator's internal truth parameters.

4.3 Decoupled Validation Holdout vs. Full-Depth Science Evaluation

We strictly separate predictive model validation from final depth measurement:

- **Exposure Holdout (Predictive Generalization Gate):** Multi-dither exposures (e.g., JWST NIRCam SMACS J0723 in F090W, F150W, F200W, F277W, F356W, F444W) are split by dither pattern (Train $= \{1, 2, 4, 5, 7\}$, Val $= \{3, 8\}$, Test $= \{6, 9\}$). This holdout is used exclusively to test out-of-fold predictive likelihood and verify that nuisance models do not overfit.

- **Full-Depth Science Evaluation:** Once hyperparameters and architecture are frozen, final scientific depth is measured using the complete exposure set across spatial-block cross-validation folds, matching real operational survey conditions.

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5 Statistical Claim Gates & Quantitative Metrics

A reconstruction is scientifically certified as *deeper* if and only if it satisfies all of the following pre-registered statistical gates:

1. **Prevalence-Independent Completeness Depth ($\Delta m_{50} > 0$):** Let m_{50} be the reference-band AB magnitude at which the fitted source recovery probability is exactly 50%, evaluated at a **fixed physical false-positive surface density** $\rho_{\text{false}} \leq \rho_0$ (e.g., $\leq 0.05\,\text{arcmin}^{-2}$ on blank/null fields), avoiding the prevalence dependence of sample-density purity:

$$\Delta m_{50} = m_{50}(\text{Continuous Latent Model}) - m_{50}(\text{Native-Pixel Baseline}). \tag{32}$$

   The paired 10,000-sample cluster-bootstrap 95% confidence interval for Δm_{50} must strictly exclude zero.

2. **Resolution-Matched Sensitivity Verification:** To prove that apparent variance reduction is not an artifact of prior over-smoothing or source broadening, we measure the 2D injection transfer function $\mathcal{H}_{\text{eff}}(\boldsymbol{x})$ and effective PSF. Depth comparisons are evaluated at **matched effective spatial resolution**:

$$\sigma_{a,\text{eff}} = \frac{\sqrt{\text{Var}(\hat{a})}}{|R_a|} < \sigma_{a,\text{eff}}^{\text{baseline}} \quad \text{at matched spatial resolution.} \tag{33}$$

3. **Photometric & Color Invariants:**
   - Bright point-source median absolute flux bias $\leq 2\%$ per band.
   - Adjacent-band median color bias $\leq 0.02\,\text{mag}$ across the smooth-continuum population.

4. **Population-Specific Expected Headroom Analysis:** Completeness curves m_{50} must be reported separately across distinct astrophysical classes, reflecting their expected headroom hierarchy (isolated point sources < extended sources < blends / structured multi-band sources):
   - *Isolated Point Sources:* Limited headroom (heuristic working target $\Delta m_{50} \sim 0.05 - 0.2\,\text{mag}$); the native-pixel GLRT is already near Neyman-Pearson optimal.
   - *Faint Extended Galaxies:* Moderate headroom (heuristic working target $\Delta m_{50} \sim 0.1 - 0.3\,\text{mag}$); multi-exposure joint modeling beats rigid template matching.
   - *Crowded Blended Pairs ($\Delta\theta \leq 1.5\,\text{FWHM}$):* **High headroom**; multi-view chromatic deblending resolves sources lost to classical confusion.
   - *Extreme Dropouts ($z > 10$) & Emission-Line Galaxies:* **High headroom**; joint cross-band evidence accumulation aggregates sub-threshold signals.

5. **Decisive Two-Stage Out-of-Sample Falsification Test (Held-Out Photon Prediction):** To conclusively distinguish genuine sensitivity improvements from prior-driven hallucinations, we enforce a two-stage discovery and confirmation protocol:

(a) *Discovery Reconstruction:* The model is fitted strictly on training exposures (Train = $\{1, 2, 4, 5, 7\}$) to identify candidate sources discovered exclusively by the continuous method.

(b) *Untouched Photon Prediction Test:* Candidate sources must statistically predict photon arrival patterns in untouched test exposures (Test = $\{6, 9\}$) under the held-out likelihood-ratio statistic:

$$\Delta \log \mathcal{L}_{\text{heldout}} = \log p(\boldsymbol{y}_{\text{test}} \mid \hat{I}_{\text{candidate}}) - \log p(\boldsymbol{y}_{\text{test}} \mid I_{\text{no-candidate}}) > \tau_{\text{null}}, \tag{34}$$

where threshold τ_{null} is calibrated on blank/null sky positions to control random false positives.

(c) *Production Refit:* Only after confirmation are all exposures combined for final science products.

6. **Spatial Homogeneity:** Depth gains must hold across at least two independent spatial quadrants, confirming that the improvement is not localized to a single favorable patch.

7. **Model Selection on Validation Batches Only:** Every (scene kind, operator rung) cell of the model zoo (§1.3, §1.6) is scored on validation batches only (Val = $\{3, 8\}$): Δm_{50} with 10,000-sample cluster-bootstrap CIs, matched-resolution $\sigma_{a,\text{eff}}$, flux/color invariants, and held-out $\Delta \log \mathcal{L}$. The winning cell is the largest Δm_{50} among cells passing all invariants; the untouched test exposures are scored exactly once after selection. A learned-PSF cell that fails the held-out-star NLL improvement gate is reported but cannot win unless it also beats the analytic arm's held-out gate.

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6 COSMOS Scale-Up Configuration

The SMACS J0723 protocol generalizes mechanically to a wide-field survey target. The reference scale-up is **COSMOS**: a $2\,\text{deg}^2$ field centered at $(\alpha, \delta) = (150.119°, +2.205°)$, combining:

- **Space UV/Optical/IR:** JWST/NIRCam COSMOS-Web F115W/F150W/F277W/F444W and MIRI F770W; HST/ACS F606W/F814W;
- **Ground Optical:** Subaru/HSC $g\,r\,i\,z\,y$; CFHT/MegaCam u^*.

The field is partitioned into $0.25\,\text{deg}^2$ tiles with 128-pixel overlap; adjacent tiles are blended by a partition-of-unity weight so that each tile carries its own `FieldDomain` while global scene continuity is preserved across seams. Each instrument contributes an independent operator bundle: learned PSFs where calibration star density allows (NIRCam, ACS), analytic fitted kernels otherwise (ground-based seeing-dominated bands) — the hybrid ladder of §1.6 is precisely what makes heterogeneous multi-instrument inference tractable. All claim gates of §5 are evaluated identically per tile.

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7 Astrophysical Applications & Discovery Potential

The ability to recover continuous, joint spatial-spectral sky fields directly from native detector exposures unlocks major scientific capabilities:

- **Cosmic Dawn & High-Redshift Dropouts ($z > 10$):** Faint primeval galaxies near the detection limit exhibit steep Lyman-break spectral cutoffs. A joint continuous spatial-spectral model borrows signal across all red/NIR channels while preserving sharp spectral discontinuities, recovering candidates that fall below the 3σ threshold in individual classical stacks.

- **Strong Gravitational Lensing & Extreme De-blending:** In rich cluster cores, highly magnified, distorted arclets are embedded within bright cluster galaxy light and diffuse intra-cluster light (ICL). Continuous representations decouple compact point-like knots from diffuse extended emission without pixelation artifacts.

- **Heterogeneous Multi-Observatory Synergy:** Seamlessly co-models observations from disparate space and ground optical/infrared imaging facilities (e.g., HST optical, JWST NIR/MIR, Euclid, Rubin LSST, and Roman), each possessing distinct pixel grids, non-uniform PSFs, and dither strategies, into a single physically consistent $(\alpha, \delta, \lambda)$ scene.

- **Discovery of Unmodeled Structure via Open Residuals:** Subtracting the forward-rendered model from native detector exposures yields calibrated *open residual maps*, exposing moving solar system objects, rapid transients, extreme emission-line sources, or exotic diffuse structures that violate smooth continuum priors.

- **Foundation Models for Cosmology (cosmodist):** Directly interfaces with next-generation deep learning architectures, providing exact unit-sphere 3D WCS coordinate mappings and continuous multi-modal cutouts for generative cosmological field modeling and distributional conformal inference.